

Multi-stage Port Scan → DoS SYN Flood from 198.51.100.54

*Investigation `inv\_01M1R4BZA0CNEN10QG9Q2NPM65` · alert `alr\_01M1R4B9AWG9QE8QQJ3QYY4KWM`
· generated 2026-09-05T06:33:50.722794+00:00*

Severity: MEDIUM

- base severity for category Reconnaissance = low
- multi-stage chain (2 supported stages) → +1

Alert summary

- _id: alr_01M1R4B9AWG9QE8QQJ3QYY4KWM
- attack_type: Port Scan
- category: Reconnaissance
- predicted_label: Recon-PortScan
- severity: low
- source_ip: 198.51.100.54
- destination_ips: ['192.168.10.34']
- device_ids: ['iot-hub-04']
- protocols: ['ARP', 'DHCP', 'DNS', 'ICMP', 'TCP', 'Telnet', 'UDP']
- event_count: 60
- confidence_mean: 0.9987191299597422
- first_seen: 2026-09-05T03:35:39.023000+00:00
- last_seen: 2026-09-05T03:36:34.494000+00:00
- model_version: 20260905T063308Z

Analyst summary

3-stage multi-stage incident from 198.51.100.54; severity medium; 4 evidence items, 9/9 claims evidence-linked.

Attack chain

1. Port Scan (Reconnaissance) — 60 flows, 2026-09-05T03:35:39.023000+00:00 → 2026-09-05T03:36:34.494000+00:00; sources 198.51.100.54; targets 192.168.10.34, iot-hub-04; support: strong; evidence: evt_01M1R4B8BA5ZH97AMY8NR006F, evt_01M1R4B8BA5ZH97AMY8NR006G, evt_01M1R4B8BA5ZH97AMY8NR006H, evt_01M1R4B8BBA315DXQ3GJ1980B3, evt_01M1R4B8BBA315DXQ3GJ1980B4
 2. DoS SYN Flood (DoS) — 219 flows, 2026-09-05T03:38:35.252000+00:00 → 2026-09-05T03:40:34.996000+00:00; sources 198.51.100.54; targets 192.168.10.34, iot-hub-04; support: strong; evidence: evt_01M1R4B8BMCMM95SY8HR161WRX, evt_01M1R4B8BNJZ9YPN7ZMEAK7M3F, evt_01M1R4B8BNJZ9YPN7ZMEAK7M3G, evt_01M1R4B8BNJZ9YPN7ZMEAK7M3H, evt_01M1R4B8BNJZ9YPN7ZMEAK7M3J
 3. DDoS SYN Flood (DDoS) — 1 flows, 2026-09-05T03:39:49.993000+00:00 → 2026-09-05T03:39:49.993000+00:00; sources 198.51.100.54; targets 192.168.10.34, iot-hub-04; support: weak; evidence: evt_01M1R4B8C8PYGP76RZCDT2RE3N
- stage 1 → 2: strong link, gap 120.758 s, shared targets 192.168.10.34, iot-hub-04
 - stage 2 → 3: strong link, gap -45.003 s, shared targets 192.168.10.34, iot-hub-04

Method: deterministic: Attack nodes PERFORMED by the alert source ordered by first_seen; other sources flooding the same target are listed as distributed_sources; other activity on the targets is context only

Evidence-backed claims

- ■ Alert alr_01M1R4B9AWG9QE8QQJ3QYY4KWM: 60 flows from 198.51.100.54 predicted as Port Scan (mean confidence 1.00) between 2026-09-05T03:35:39.023000+00:00 and 2026-09-05T03:36:34.494000+00:00.

evidence: evt_01M1R4B8BA5ZH97AMY8NR006F, evt_01M1R4B8BA5ZH97AMY8NR006G, evt_01M1R4B8BA5ZH97AMY8NR006H, evt_01M1R4B8BBA315DXQ3GJ1980B3, evt_01M1R4B8BBA315DXQ3GJ1980B4, evt_01M1R4B8BBA315DXQ3GJ1980B5

- Stage 1: Port Scan from 198.51.100.54 against 192.168.10.34, iot-hub-04 (60 flows, 2026-09-05T03:35:39.023000+00:00 → 2026-09-05T03:36:34.494000+00:00).
evidence: evt_01M1R4B8BA5ZH97AMY8NR006F, evt_01M1R4B8BA5ZH97AMY8NR006G, evt_01M1R4B8BA5ZH97AMY8NR006H, evt_01M1R4B8BBA315DXQ3GJ1980B3, evt_01M1R4B8BBA315DXQ3GJ1980B4, evt_01M1R4B8BBA315DXQ3GJ1980B5
- Stage 2: DoS SYN Flood from 198.51.100.54 against 192.168.10.34, iot-hub-04 (219 flows, 2026-09-05T03:38:35.252000+00:00 → 2026-09-05T03:40:34.996000+00:00).
evidence: evt_01M1R4B8BMCMM95SY8HR161WRX, evt_01M1R4B8BNJZ9YPN7ZMEAK7M3F, evt_01M1R4B8BNJZ9YPN7ZMEAK7M3G, evt_01M1R4B8BNJZ9YPN7ZMEAK7M3H, evt_01M1R4B8BNJZ9YPN7ZMEAK7M3J, evt_01M1R4B8BNJZ9YPN7ZMEAK7M3K
- Stage 3: DDoS SYN Flood from 198.51.100.54 against 192.168.10.34, iot-hub-04 (1 flows, 2026-09-05T03:39:49.993000+00:00 → 2026-09-05T03:39:49.993000+00:00).
evidence: evt_01M1R4B8C8PYGP76RZCDT2RE3N
- Stage 1 preceded stage 2 by 121s (strong link: shared sources ['198.51.100.54'], shared targets ['192.168.10.34', 'iot-hub-04']).
evidence: evt_01M1R4B8BA5ZH97AMY8NR006F, evt_01M1R4B8BA5ZH97AMY8NR006G, evt_01M1R4B8BA5ZH97AMY8NR006H, evt_01M1R4B8BBA315DXQ3GJ1980B3, evt_01M1R4B8BBA315DXQ3GJ1980B4, evt_01M1R4B8BMCMM95SY8HR161WRX
- Stage 2 preceded stage 3 by -45s (strong link: shared sources ['198.51.100.54'], shared targets ['192.168.10.34', 'iot-hub-04']).
evidence: evt_01M1R4B8BMCMM95SY8HR161WRX, evt_01M1R4B8BNJZ9YPN7ZMEAK7M3F, evt_01M1R4B8BNJZ9YPN7ZMEAK7M3G, evt_01M1R4B8BNJZ9YPN7ZMEAK7M3H, evt_01M1R4B8BNJZ9YPN7ZMEAK7M3J, evt_01M1R4B8C8PYGP76RZCDT2RE3N
- get_related_events returned 220 records (predicted types: DoS SYN Flood×219, DDoS SYN Flood×1; 1 destination(s)).
evidence: evd_01M1R4BZS74NV391236AM4ZWW4
- search_alerts returned 2 records (predicted types: DoS SYN Flood×1, DDoS SYN Flood×1; 2 related alert(s): DDoS SYN Flood, DoS SYN Flood).
evidence: evd_01M1R4BZSS5J87NTKKB9Z33C45
- search_security_events returned 280 records (predicted types: DoS SYN Flood×219, Port Scan×60, DDoS SYN Flood×1; 1 destination(s)).
evidence: evd_01M1R4BZV860MM77ZCE3414XRQ

AI attack narrative

Generated by: template (LLM unavailable or narrative failed grounding validation) · model: n/a · grounding validation: {"checked": false}

At 2026-09-05T03:35:39.023000+00:00, 60 network flows from 198.51.100.54 towards 192.168.10.34 were classified as Port Scan (Reconnaissance) with mean model confidence 1.00. The flow-level features Max, Protocol Type, AVG provide behavioral evidence associated with the labelled Port Scan class [evt_01M1R4B8BA5ZH97AMY8NR006F, evt_01M1R4B8BA5ZH97AMY8NR006G, evt_01M1R4B8BA5ZH97AMY8NR006H].

Adaptive evidence collection linked the alert to related activity of the same source: stage 1 Port Scan (60 flows, 2026-09-05T03:35:39.023000+00:00 → 2026-09-05T03:36:34.494000+00:00) [evt_01M1R4B8BA5ZH97AMY8NR006F, evt_01M1R4B8BA5ZH97AMY8NR006G]; stage 2 DoS SYN Flood (219 flows, 2026-09-05T03:38:35.252000+00:00 → 2026-09-05T03:40:34.996000+00:00) [evt_01M1R4B8BMCMM95SY8HR161WRX, evt_01M1R4B8BNJZ9YPN7ZMEAK7M3F]; stage 3 DDoS SYN Flood (1 flows, weak support, 2026-09-05T03:39:49.993000+00:00 → 2026-09-05T03:39:49.993000+00:00) [evt_01M1R4B8C8PYGP76RZCDT2RE3N]. Stage order is derived from flow timestamps; stages 1→2 are a strong link (shared targets: 192.168.10.34, iot-hub-04); stages 2→3 are a strong link (shared targets: 192.168.10.34, iot-hub-04).

Threat intelligence: Insufficient evidence — no external reputation data was obtained for the source indicator (provider not configured or unavailable).

What cannot be concluded: usernames, passwords, commands, payloads, files or process activity — Insufficient evidence from network-flow features alone.

Threat intelligence (actual API results)

Insufficient evidence — no external reputation data obtained.

MITRE ATT&CK mapping

Behavioral evidence

- Max: z vs benign = -1.0218006082380149
- Protocol Type: z vs benign = -0.4889221199339297
- AVG: z vs benign = -1.0734406849902163
- Tot size: z vs benign = -1.0274880143493275
- Srate: z vs benign = 0.7168491331182526
- Rate: z vs benign = 0.7168491331182526

Supporting evidence items

- `evd\_01M1R4BZQXHDXB6GY5T73X9NT5` step 1 get_event_aggregates → 3 distinct attack_type values; top: DoS SYN Flood×219, Port Scan×60, DDoS SYN Flood×1 (records 0/280)
- `evd\_01M1R4BZS74NV391236AM4ZWW4` step 2 get_related_events → predicted types: DoS SYN Flood×219, DDoS SYN Flood×1; 1 destination(s) (records 220/220)
- `evd\_01M1R4BZSS5J87NTKKB9Z33C45` step 3 search_alerts → predicted types: DoS SYN Flood×1, DDoS SYN Flood×1; 2 related alert(s): DDoS SYN Flood, DoS SYN Flood (records 2/2)
- `evd\_01M1R4BZV860MM77ZCE3414XRQ` step 4 search_security_events → predicted types: DoS SYN Flood×219, Port Scan×60, DDoS SYN Flood×1; 1 destination(s) (records 280/280)

Knowledge graph summary

```
{
  "node_count": 13,
  "edge_count": 25,
  "by_type": {
    "Alert": 3,
    "EvidenceSet": 2,
    "IP": 2,
    "Device": 1,
    "Attack": 3,
    "Investigation": 1,
    "Behavior": 1
  },
  "last_updated_step": 4,
  "truncated": false
}
```

Recommended actions

- Block or rate-limit the single source IP at the perimeter and confirm the targeted service has recovered.
- Check whether the source performed reconnaissance earlier (precursor evidence) and monitor for repetition.
- Monitor the source for follow-on activity against the scanned devices; scanning frequently precedes exploitation.
- Review exposed services on the scanned devices and close unnecessary ports.
- Apply rate limiting / upstream filtering for the flood protocol at the gateway serving the targeted device.
- Validate service availability of the targeted device and review its exposure to the source network.

Investigation limitations

- DEMONSTRATION DATA: the flows behind this report are synthetic placeholders generated because no CICIOT2023 files were present in DATASET_DIR. No statement in this report describes the real dataset.

- Entity and time context (IP addresses, devices, timestamps) is synthesized for CICIoT2023 flows; feature values and attack labels come from the data source recorded in each event.
- Flow features are behavioral evidence associated with the labelled class; usernames, commands, payloads, files and process activity are not available from the current evidence.
- Low-support stage(s) 3 DDoS SYN Flood (1 flow): fewer than 3 flows were classified with this label; this may be a model misclassification and is reported with a weak-support flag rather than asserted as an attack step.
- No external threat-intelligence data was obtained (provider not configured, indicator private/documentation range, or unavailable).

Investigation metrics

```
{
  "steps": 4,
  "db_queries": 4,
  "events_retrieved": 502,
  "llm_calls": 0,
  "ti_lookups": 0,
  "evidence_items": 4,
  "claims": 9,
  "claims_validated": 9
}
```